

Institute of Actuaries of India

Subject SP5 –Investment and Finance

May 2023 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) The key aims behind developing International Financial Reporting Standards (IFRS) are as follows:
- Encouraging reliable and consistent accounting data [1]
 - Transparency of accounting data [0.5]
 - Single set of standards world-wide enabling access to financial markets. [0.5]
 - Prevention of companies having to produce results based on biases. [0.5]
 - Making it easier for companies to obtain listing on different stock exchanges as one set of standards become acceptable to all listing authorities in major financial centres. [1]
 - Mitigation of issues faced in past economic and financial crises due to a perceived lack of transparent and reliable financial information [1]
 - Helping investors make informed choices, reducing their risks [1]
 - Reducing cost of capital for companies by providing easier confidence to investors. [0.5]
- [Max 6]**
- ii) The key requirements of IFRS 9 are:
- Showing instruments at fair value in the statement of financial position. [1]
 - Showing any resultant profit or loss on revaluation in the statement of comprehensive income. [1]
- [2]**
- iii) Setting up a market risk monitoring and control system in the company would involve the following steps:
- Determine a suitable set of measures to track market risk exposure (e.g. Value at Risk, deviation from benchmark asset distribution)
 - Setting up a reporting system so that these metrics are tracked on a regular basis
 - Agree on the frequency of reporting with company management
 - Automation of standard data input processes so that changes in asset values or portfolio mix are captured quickly
 - Ability to measure and track the impact of proposed management actions
 - The system should be simple enough so that users (such as company staff or fund managers) can understand and operate it
 - The monitoring and risk management should be done by personnel independent of the fund managers
 - The output should be quantifiable
 - The thresholds for quantitative outputs, for e.g. risk limits based on VaR or maximum deviation should be agreed with management
 - Major deviations should have a standard set of proposed mitigation actions in case the risks materialize or breach thresholds.
- [0.5 mark for each point, **Max 5]**
- iv) The success of a government's economic policy can be measured by the following key indicators:
- Unemployment
 - Inflation
 - Balance of payments
 - Economic growth.
- [0.5 mark for each point, **Max 2]**
- v) Mean-variance portfolio theory, sometimes called modern portfolio theory, specifies a method for an investor to construct a portfolio that gives the maximum expected return for a specified level of risk... [1]
 ... or the minimum risk for a specified expected return. [0.5]
 However, the theory relies on some strong and limiting assumptions about the properties of portfolios that are important to investors. [1]
- [2]**
- vi) The key assumptions of mean-variance portfolio theory are:

- All expected returns, variances and covariances of pairs of assets are known. [1]
- Investors make their decisions purely on the basis of expected return or variance [1]
- Investors are non-satiated [0.5]
- Investors are risk-averse [0.5]
- There is a fixed single-step time period [0.5]
- There are no taxes or transaction costs [0.5]
- Assets may be held in any amounts, i.e. short-selling is possible, we can have infinitely divisible holdings, and there are no maximum investment limits. [1]

[Max 4]**[21 Marks]****Solution 2:****i) Futures**

- You would need to find a domestic market equity index future that closely matches the portfolio of the company you have joined. The equity exposure can be reduced by selling equity futures. [1]
- You could buy international market equity index futures at the same time as selling the domestic equity index futures. [0.5]

The relative advantages and disadvantages of using futures would be:

- It could be put in place quickly given the liquidity of futures markets. [0.5]
- It would be relatively inexpensive assuming mainstream equity future contracts (not thinly traded ones) are used. [1]
- Contracts would need to be rolled over, incurring costs and risks. [0.5]
- Potential exposure to cross hedging, as equities underlying futures are likely to differ from actual equities held. [0.5]

Options

- The exposure would be calculated as delta * nominal exposure. [0.5]
- Purchasing call options on the overseas market to ensure the equity exposure is sufficient would be the simplest way. [0.5]
- Call options on the domestic equity market exposure could be sold to hedge domestic exposure. [0.5]

Advantages and disadvantages include:

- This route would make it easier to get exposure to upside from international markets without the downside risk. [1]
- Put options could be purchased in the domestic market, retaining upside potential while hedging downside risks. [1]
- On the other hand, options are likely to be less liquid than futures contracts. [0.5]
- If delta hedging is used, this would need constant monitoring and rebalancing. [0.5]
- Significant premium income may be required for some option strategies, compared to futures where only the margin is required. [0.5]

Swaps

- This would require a contract between you and another investor, where the other investor passes the proceeds from the national domestic equity index portfolio in return for the returns from an international equity index portfolio over an agreed period of time. [1]
- The size and nature of the contract could be flexible and could be tweaked to suit both parties. [0.5]
- The cost may be lower as the middle man has been cut out... [0.5]
- ... except a possible small fee for a bank making the deal. [0.5]
- The term of the deal would not be restricted by the availability of appropriate futures or options contracts with the right expiry dates. [1]

- The deal can be reversed at any time if agreed upon by both parties, or reversed by a future offsetting deal. [1]

[Max 10 Marks]

ii)

- Any model for asset prices is likely to have limitations, regardless of the distribution, characteristics and expert judgement used. [0.5]
- Several mathematical techniques can be used to mitigate the shortcoming of your existing models such as non-constant parameters, fatter tails or non-independent increments. [0.5]
- Studies of equity returns, interest rates and foreign exchange rates have shown significant non-linear dependence, with absolute values of returns and squares of returns significantly autocorrelated. This can be considered in the choice of any model or technique. [1]

Alternate distributions

- You can start by considering alternate probability distributions. [0.5]
- While the normal distribution is commonly used to model asset prices, there are other distributions such as the student's t-distribution. [0.5]
- These other distributions can account for the fatter tails and other non-normal characteristics of financial data. [0.5]

Time series techniques

- You could use time-series analysis techniques to aid your modelling. [0.5]
- Time-series analyses could be appropriate for stock prices, as it is a technique used to model and analyse time-dependent data. [0.5]
- A heteroscedastic model could be used, in which the volatility of the time series being modelled varies in some way. [0.5]
- For example, in an autoregressive conditional heteroscedastic model, the volatility parameter is modelled as a time-varying process in its own right. [0.5]

Volatility parameter

- Studies of stock market volatility over long periods suggest that market volatility varies in systematic ways. [0.5]
- For example, volatility increases in recessions and is lower in bull markets. [0.5]
- Markets also exhibit volatility "clustering", where large and small shocks tend to be clustered together. [0.5]
- Accurate modelling of the above features is likely to be difficult to develop. [0.5]

Additional factors

- Additional factors impacting the market price, such as market trends, company performance, and macroeconomic conditions, should be factored into your model to improve accuracy. [0.5]
- The relationship between key parameters such as equity values, dividend growth and inflation could be analysed using time series techniques. [0.5]
- These studies should be interpreted with care to determine whether and how equity markets and inflation indices are correlated. [1]

Other techniques

- Stochastic modelling could be a powerful tool to model financial data, including asset prices. [0.5]
- Advanced techniques available to your company, such as the use of machine learning or AI algorithms, could also be explored to enhance your statistical models. [0.5]

[Max 10]

iii) If you expect the Actuarial stock market to perform poorly for the next six months and you would like to utilise futures, this would be the process:

- Sell a six-month futures contract reflecting your expectations about the stock market. [1]
- This may be better than dealing in the cash market because:

- Your initial capital outlay is only the initial margin, so you can heavily gear up your investment subject to the risk of losses. [1]
- Administratively easier than dealing in the cash market. [0.5]
- Lower expenses in dealing in futures in most cases [0.5]
- There is no need to worry about stock selection, which can save time and costs. [1]

[4]

[24 Marks]

Solution 3: There are a number of potential practical problems with this approach, including:

- This approach requires the bonds to be held equal in present value to the discounted value of the future rental payments. [0.5]
- The company may not have funds available to invest in bonds at the required time to match the rent. [0.5]
- The difficulties will be higher from smaller companies with less liquid or diverse asset holdings. [0.5]
- The use of zero coupon bonds is likely to give rise to credit risk, which is not matched by the liabilities. [0.5]
- Bonds of the appropriate term, especially long enough to meet the rental payments, may not be available. [0.5]
- If bonds of a shorter duration are held, this will give rise to reinvestment risk. [0.5]
- Depending on the country's local investment market and any restrictions, additional issues may arise in the purchase of assets. [0.5]
- Bond maturities may not exactly match the timing of rental payments. [0.5]
- For example, rent may be due on a monthly basis but bonds may only be available with term options for every 3 months or 6 months. [0.5]
- The tax status of the bonds will also have an impact. [0.5]
- In particular, the tax cash flows for assets and liabilities (rental payments) are unlikely to match. [0.5]

[Max 5 Marks]

Solution 4:

- Overconfidence & optimism
- Representation bias – putting too much emphasis on Thai official's beliefs
- Belief preservation – did not alter his fund position even with new information alarming otherwise
- Prospect theory – gambling at a higher stake & wanting to make up for his poor return in recent past
- Noise trading – getting swayed too much by what's "hot" rather than finding opportunities based on solid research

[5 Marks]

Solution 5:

- Not dynamic, stresses one variable and keeps everything else constant.
- Model risk when simplifying a complex real life situation into a simplistic models into a few variables. Possibility of structuring an incorrect model.
- Parametric risk introduced when incorrect parameters or distribution of parameters is selected to run the resilience test.
- Complexity in building the model & handling correlations
- Resources in cost of hiring expertise & time involved in building the model

[5 Marks]

Solution 6:

- Mr.A's son's education expenses have doubled and had an unplanned vacation abroad, but has not changed his asset liability strategy since benchmarking it at the start of the year, even though his liabilities have significantly changed during the course of the year.

- Mr.A should have used dynamic liability benchmarking, which are best suited to circumstances where the typical benchmark needs to vary based on the changing nature of liabilities.
- Benchmarks will always need to be reviewed periodically to ensure they remain appropriate, but where the liabilities are changing rapidly, then it may be necessary to link the benchmarks more closely to the liability portfolio at all times.
- Dynamic liability benchmarks represent an intermediate portion between static benchmarks, which are reviewed infrequently, and full liability hedging which are structured in a way that assets behave in the same way as liabilities.
- Dynamic liability benchmarks are reviewed much more frequently, perhaps weekly or monthly.

[5 Marks]

Solution 7:

- i) Securities Transaction Tax (STT) is a tax payable in India on the value of securities transacted through a recognized stock exchange [1 mark]

Implications:

- Lower liquidity
- Higher bid-ask
- Higher revenue for government
- Reduced high frequency trading
- Shift of trade holding to longer durations

[4]

ii)

- Withholding taxes, also called tax deducted at source, are a said amount of taxes that are withheld from payments made
- The amount withheld is taken as a credit to the person's income tax payable value
- If too much money is withheld, an employee receives a tax refund or may have to pay the tax authority if not enough is withheld.

[3]

[7 Marks]

Solution 8:

- i) For: markets tend to over react or under react creating opportunities, easier to collect data, quick and easy to carry out, can be useful for timing of investment

Against: past data may not repeat itself, efficient market hypothesis, increased active trading resulting in more expenses, not relying on long term value

[2]

- ii) 20 day moving average falling below 40 day moving average implies:

- The stock is in a downward trend
- Is a bearish sign and, in the absence of other information, indicates the stock is expected to fall further.

[1 mark for each valid point, **Max 2**]

- iii) Contrarian & momentum. Momentum outperforms during periods of boom while contrarian outperforms during periods of recession.

[0.5 marks for identifying the strategy, 1 mark for each explanation]

[3]

iv)

- a) Basis risk, when it is not always possible to find futures contracts based on the underlying that trade in the market
- b) Cross hedging risk, when when it may be impossible to find futures contracts on a particular underlying asset

[1 mark each, **Max 2]****[9 Marks]****Solution 9:**

i) & ii)

	FNN	FAA	Asset Alloc	Stock Sel	Total
Index Linked Bonds	16.77%	19.01%	1.92%	0.56%	2.48%
Domestic Equities	17.82%	25.45%	0.00%	1.91%	1.91%
Overseas Equities	46.38%	40.71%	5.49%	-2.83%	2.65%
Total	24.43%	31.47%	7.40%	-0.36%	7.04%

FNN	Compounded Benchmark Return of Index with Strategic Allocation
FAA	Compounded Benchmark Return of Index with Actual Allocation
Asset Alloc	$(\text{Actual Allocation} - \text{Strategic Allocation}) * (\text{Benchmark Asset Return} - \text{Overall Benchmark Return})$
Stock Alloc.	$\text{Actual Allocation} * (\text{FAA} - \text{FNN})$

[1 mark each for correct FNN & FNA – **Max 2]**[.5 marks each for subheading of Asset Alloc, Stock Sel & Total entries – **Max 6]**

- iii) Outperformance primarily seems to come from asset allocation & deviating from benchmark rather than stock selection

This could turn out to be a risky strategy if the benchmark is liability hedged

The manager should look at shifting to passive investment management from active to minimize expenses, since stock selection skill is not noticable.

[2]**[10 Marks]****Solution 10:**

i)

Period	TWRR
1	-35.15%
2	25.90%
3	6.12%
4	19.99%
Total	3.96%

[1 mark for approach, .5 marks each for correct yearly calculation – **Max 3]**

ii)

Period	Investment
0	5388
1	120
2	-90
3	499
4	-129

Return	6.50%
Total Value	6,149

[1 mark for approach, .5 marks each for correct yearly calculation – **Max 3**]

iii)

Period	Return
0	-2.51%
1	4.85%
2	6.55%
3	-2.43%
4	6.27%

$$R(t) = ((I(t) * (1 + Y(t)/4)) / I(t-1) - 1) * 100$$

where $R(t)$ = total return for period t

$I(t)$ = index value at time t

$Y(t)$ = yield on index at time t

[1 mark for approach, .5 marks each for correct yearly calculation – **Max 3**]

[9 Marks]
